How Feed Water Heater Helps the Engineer

The first question that naturally occurs to the locomotive engineer, when some new device is applied to the locomotive he is running, is—"What does this so-called fuel saving or efficiency increasing appliance mean to me personally?" "Is it actually going to help me get over the division on time with regularly scheduled trains, or in less time on freight runs and with 'extras'?" "Will it prove of real assistance to me in making the grades and in reaching close meeting points with a few minutes to spare, even when hauling more than the usual number of cars?" "Is it going to be so easy to operate that it will not distract my attention from other more important duties—and will it be free from failures and troubles on the road?" The real value of the feed water heater to the engineer is readily appreciated by applying this list of questions to this particular device.

Getting Over the Road

"How is this device going to help you get over the road more easily?" It will do this, primarily, by increasing the steam generating capacity of the locomotive. To make this point clear it should be explained that where an injector is used to supply the boiler, the temperature of the feed water is raised many degrees by live steam taken directly from the boiler. This means that part of the fuel burned in the firebox goes to generate this steam which is used to heat the water passing through the injector.

The purpose of the feed water heater is to reclaim a portion of the heat of the exhaust steam from the locomotive cylinders which would otherwise be lost, and use this heat to raise the temperature of the boiler feed water. It is plainly seen that in doing this the feed water heater relieves the boiler of a certain amount of work necessary if an injector were used, and this materially increases the steaming capacity of the locomotive boiler for the same amount of fuel burned.

For example, in the case of a locomotive with 200 lbs. steam pressure, 150 degrees superheat and 40 degrees tender water temperature, the claim is made that the Worthington feed water heater will increase the steaming capacity 13.5 to 14.5 per cent depending on exhaust steam pressure.

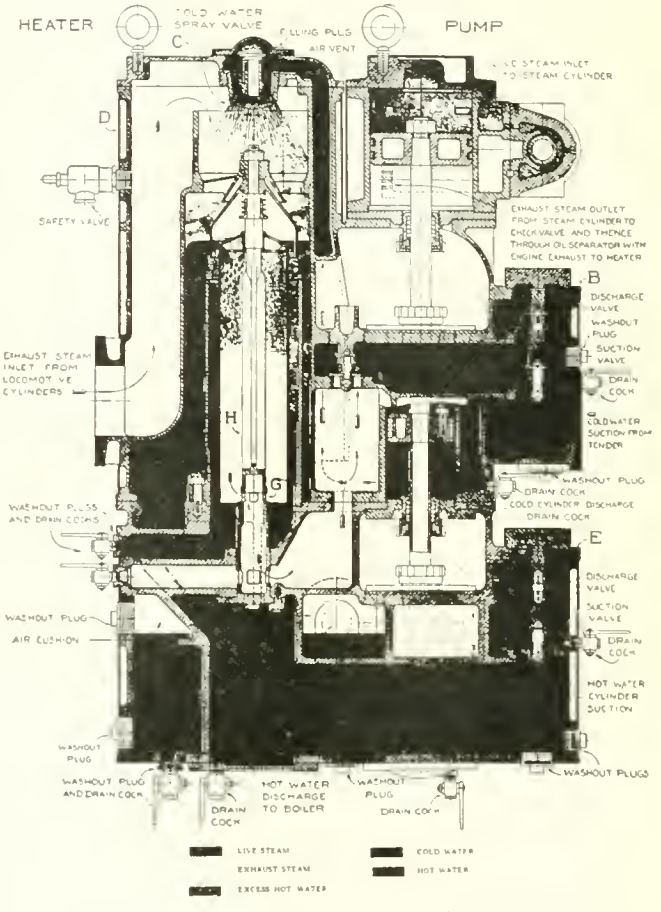
This increase in locomotive steaming capacity can, perhaps, be more clearly shown as follows: A Mikado type locomotive which evaporates on an average of 6¾ lbs. of water per pound of coal with an injector, can evaporate 27,000 lbs. of water with 4,000 lbs. of coal per hour. The same locomotive when equipped with a Worthington feed water heater will evaporate approximately 29,900 lbs. of water with the same amount of coal.

There is still another way in which the Worthington feed water heater is of great assistance to you in getting over the road on time and in less time, and that is because of the fact that it is an open heater. The exhaust steam condensed in heating the feed water mixes directly with the water taken from the tank. This means that the heater will recover from the exhaust steam from 13 to 16 per cent of the total water supplied to the boiler. This water saving is equivalent to a considerable increase in tender water capacity, and enables you to pass more water plugs on the road without stopping to fill the tank.

From the foregoing it is apparent that the feed water heater will prove a vital factor in helping you get over the road, by decisively increasing the steaming capacity of the locomotive, and materially reducing the amount of water used on the road.

Making Grades and Meeting Points

How does the feed water heater assist you in making the hill or help you in "hitting her up" to reach a meeting point on time? It does this by increasing the steam generating capacity of the boiler at the very time when you are working the engine hardest and need this help most. This is a peculiar advantage offered by the feed water heater. When the locomotive is working the hardest on heavy grades, or in starting trains, the increased amount of heat furnished by the exhaust to the feed water is of great assistance at just the time when boiler capacity is taxed to the utmost. Also in extremely cold winter weather, when trains are difficult to handle, the feed water heater comes to the rescue by utilizing the waste heat of the exhaust to maximum advantage in raising the temperature of the extremely cold water supplied from the tank, thus assuring more freely steaming locomotives just at the critical time when they are most needed.



The Worthington Feed Water Heater

Ease of Operation

Coming to the third question—"Is the feed water heater easy to operate?" Yes. It is much easier to operate than an injector, as a slight turn or adjustment of the pump throttle valve is all that is needed to maintain a constant water level under all operating conditions. Wherever the Worthington is used, engine crews invariably prefer it to the injector. It can be operated continuously at low capacities and can be kept in operation without attention from engine crew, even when the steam pressure is very low. It is not necessary for the engineer to take his attention

at either end, all enclosed in a cast iron body. The water passes through the tubes over the agitators placed in each tube and continues directly from the heater to the boiler check. In this heater there are no working parts or loose pieces. The condensate is conducted from the bottom of the heater to a filter located on the tender, where the oil is separated and the filtered and distilled water is discharged to the tank.

Open Type Heater

The open type heater consists of a combined heater and pump, so arranged that one steam cylinder drives two water pistons in separate cylinders, one of which draws cold water from the tender and discharges it through a spray valve in the top of the adjacent heater, and the other takes the hot water from the bottom of the heater and forces it into the boiler. In this type of heater the condensate is added to the feed water in the heater and since the quantity of this varies, it is necessary to incorporate within the heater a control device consisting of a large bucket type float and by-pass passages and valves to prevent the heater filling up in case there is formed more hot water than the lower pump can discharge. The oil is removed from the exhaust steam by a separator before it reaches the heater.

Exhaust Steam Injector

The exhaust steam injector, or the injector type open feed water heater, as the name implies, not only feeds the boiler by the use of exhaust steam, but delivers the feed water at temperatures varying from 200 to 220 degrees. Its operation is entirely automatic, as it operates with exhaust steam when the latter is available, and with live steam when the throttle is closed, without manipulation on the part of the enginemen. This feature of its operation is controlled by a differential piston, having its connection to the main steam pipe of the locomotive, steam pressure holding the live steam valve closed when the locomotive is using steam and when exhaust steam is available for working the injector.

The working of the injector is extremely simple, when instructions for its operation are understood, with either exhaust steam or live steam, and with the improved operating mechanism, no special instructions are required by enginemen, who are familiar with working any of the live steam injectors now in general use, as the operating parts are confined to a regulator and one lever, having four positions, namely; closed, priming, automatic and positive live steam position. The positive live steam position may be considered an emergency feature, since it is only required to operate the injector when the locomotive is standing with a pressure in the cylinders. An oil separator is used in the exhaust steam line for removing oil and other deposits in the exhaust steam before it reaches the injector.

The efficiency of the exhaust steam injector as a feed water heater, its simplicity of construction, flexibility and dependability, either when used as a feed water heater or a live steam injector, make it a very attractive means of boiler feeding. The fact that it will put water in the boiler against high pressures with less than 1 pound pressure, is one of its remarkable features.

When the locomotive is using steam and the injector operating with exhaust steam, it is essentially an open type heater, as the exhaust steam condenses in the water similar to other types of open heaters and the condensate is returned to the boiler; therefore, the economies that may be expected from its use are the same as with other heaters of this type.

While the exhaust steam injector for locomotive service is comparatively new in this country, there are a great many locomotives equipped with them in foreign countries.

It will be seen that the application of a feed water heater to a locomotive results in three principal advantages. First, it reduces the amount of coal required to produce the required amount of power; second, it increases the available water supply and, third, it reduces the amount of back pressure on the pistons, making the locomotive easier in action and more powerful. These are all very real and practical gains which all traveling engineers can fully endorse.

When coal costs from $40.00 up to possibly $125.00 for each tender load, a saving of about 15 per cent represents a very substantial amount of money. In most cases feed water heaters are applied chiefly for this reason. In addition to the saving in the cost of the coal consumed, there is also the reduction in the quantity of company coal to be hauled and handled, which in severe weather or periods of heavy traffic, is an item of some importance.

Occasionally it happens, especially on very heavy passenger power, that the gain in quantity of water available is of equal or greater value than the reduction in fuel consumption. No one ever heard an engineer or despatcher admit that the tender of an engine carried too much water. A feed water heater by condensing exhaust steam and returning it to the tender, saves about 1,500 gallons in 10,000 gallons, without increasing its size or weight. It sometimes costs a surprising amount of money to make an unnecesary stop with a heavy train and the advantage offered by the heater in this direction is not to be overlooked.

Tests

There have been a large number of service tests made with feed water heaters during the past few years. These have covered both passenger and freight service on both mountainous and level districts, and in general all show practically the same results. The following are some typical examples of the results of such tests:

	With Heater	Without Heater	%	With Heater	Without Heater	%	With Heater	Without Heater	%
Type locomotive..	4-6-2	4-6-2	...	2-8-0	2-8-0	...	2-8-2	2-8-2	...
Class service.....	Pass.	Pass.	...	Fr't	Fr't	...	Fr't	Fr't	...
Tonnage train ...	481	506.5	5	2,759	2,724	1.3	3,251	3,224	.8
Running time	3-3	3.23	...	6.8	6.13	...	5.12	5.35	6.5
Av. steam pressure.	201.6	202	...	203.5	203	...	173	172.5	...
Total fuel—lbs ..	9,617	11,000	12.6	22,641	26,370	14.1	18,393	21,640	15
Water evaporated per pounds fuel burned	7.82	6.75	13.7	7.2	6	20	7.807	6.698	16.55

A year or so ago one railroad made a very careful investigation to determine the exact gain made by the reduction in back pressure on the pistons by the application of feed water heaters. A series of comparative runs were made where conditions of speed, cut-off and steam pressure were exactly the same at certain points on the road. Indicator cards were taken at these points, and a comparison developed the fact that 3¼ per cent less steam was required to give a unit of work with the heater than when using the live steam injector at a speed of 10 miles an hour, and over 6 per cent at from 35 to 43 miles an hour. This was all due to the reduction in back pressure caused by the condensing action of the heater.

Another more crude but no less convincing example of the same effect occurred on some comparative runs with and without heaters in use on a heavy passenger locomotive. On one portion of the run there was a very heavy grade about 15 miles long. It was found that when the heater was used, the same weight train could be taken up this grade in from two to three minutes less time.

These three chief advantages of a feed water heater on a locomotive are no longer a subject for debate or discussion. There are now well over a thousand feed water heaters in service in this country, and many road or service tests made under all possible conditions have proved them

an hour by making 5, 10 and 15 lbs. reductions and then placing the engineer's valve in lap position in each case until the train stops.

After this has been done the train will be switched or the piston travel adjusted so as to have piston travel of 4 to 6 inches on first 18 cars and last 18 cars, and 9 inches or more on remainder of A. S. A. cars. On Westinghouse cars, piston travel to be normal. The above outlined tests will then be repeated.

The running tests of Series C will be made from Creive to Roanoke, 128 miles, with an empty train of 75 cars all equipped with the A. S. A. brake. On this run, brake tests will be made at suitable locations, and notes will be made as to whether the shocks which occur, if any, are unusual or excessive. The piston travel on first 19 cars and last 19 cars to be 9 inches or more, and on the remaining 37 cars 4 to 6 inches. Brake applications and stops will be made as before from 10, 15 and 20 miles an hour, and in addition thereto, an emergency stop will be made from 15 miles an hour and another from 7 miles an hour. After this last test the train will be switched and the piston travel will be adjusted so as to have piston travel 4 to 6 inches on the first 17 cars and the last 21 cars, and 9 inches or more on the intermediate 37 cars.

The above schedule of tests will then be repeated.

On arrival at Roanoke the Series D of the standing tests will be made. In these tests the piston travel on the first 17 and last 21 cars is to be from 4 to 6 in. and 9 in. or more on the remainder of the cars.

Reductions of 5, 10 and 15 lbs. will then be made and after each reduction the brake valve will be placed on lap and left in that position until the conclusion of the test. A note will be made of the brake cylinder and brakepipe pressure on each car that is equipped with gauges each minute for 3 minutes after the brake starts to apply.

The train will then be switched so as to have the piston travel on the first 18 and last 19 cars, 9 inches and from 4 in. to 6 in. on the remaining cars of the train. The above outlined test will then be repeated.

In the Series E tests, the piston travel will be made 9 in. or more on each car. First there will be a 3 lb. reduction, and the brake valve placed in lap position where it will be left until the conclusion of the test, and the same memoranda made as in the case of the D Series tests. Then the test will be repeated with a 5 lb. and 8 lb. reduction successively.

In the case of the F Series, the piston travel will be adjusted on each car so that it will be between 4 in. and 6 in. and the tests of Series D repeated.

In the Series G tests the piston travel will be so adjusted as to lie between 6 in. and 8 in. on each car, and the tests of Series D repeated.

In the Series H tests the normal piston travel of from 6 in. to 8 in. will be used on each car. First the brake cylinder leakage on each car will be noted and it will be adjusted to approximately 10 lbs. per minute from a pressure of 50 pounds.

First there is to be an 8 lb. reduction and the brake valve placed in lap position. The brake cylinder and brakepipe pressures will be taken on each car equipped with gauges for 5 minutes after the brake starts to apply. The brake will then be released and a note made of the time required to release and recharge to the initial pressure.

Then with a 20 lb. reduction the brake valve will be put on lap and held for one minute after brakepipe exhaust ceases. Then release and recharge as before.

These two tests will then be repeated with all of the triple valves in graduated release.

Then the brakes will be adjusted to operate in quick release, when an emergency application will be made, leaving the brake valve in emergency position, and note whether any brakes release in a period of 5 minutes. Brake cylinder leakage on each car is then to be adjusted to approximately 17 lbs. per minute from 50 lbs., and the above tests repeated.

The same is to be done with the brake cylinder leakage on car 49 to be created equivalent to amount of air which can be supplied by the triple valve. Brake cylinder leakage on all other cars to be normal. This last condition is to be created first on cars 39 and 49, and then on cars 29, 39 and 49, with the leakage on the other cars normal and the tests repeated. Finally this same amount of leakage is to be created on car 2, that on all other cars remaining normal, and the tests repeated.

The work will then go back to Series B and a running test will be made with a train of 75 empty cars from Bluefield to Roanoke, 106 miles, in order to demonstrate the control of trains of empty cars having Westinghouse and A. S. A. equipment mixed. In this train the first 25 cars next to the engine will be equipped with Westinghouse triple valves, the next 25 cars will have the A. S. A. triple valves, and the last 25 cars will have Westinghouse valves. Running brake tests will be made at suitable locations en route, noting whether the shocks which occur, if any, are unusual or excessive and whether the operation of either type of triple valve has any effect, either adverse or beneficial, upon the operation of the other type of triple valve.

The piston travel will be adjusted to 6 in. to 8 in. in accordance with the practice of the Norfolk & Western Railway Company.

On this run the tests outlined for the empty car train in the other Series B tests between Norfolk and Creive, will be repeated.

A final test of Series I will be made with a loaded train of 50 cars, from Bluefield to Roanoke to demonstrate the control of a loaded train having Westinghouse and A. S. A. equipment mixed. The train will be made up as follows: 10 cars equipped with Westinghouse triple valves, 10 cars equipped with A. S. A. triple valves, 10 cars equipped with Westinghouse triple valves, 10 cars equipped with A. S. A. triple valves, 5 cars equipped with Westinghouse triple valves and 5 cars equipped with A. S. A. triple valves; all retainers on Westinghouse cars to be turned up, and all A. S. A. triple valves to be operated in graduated release.

This train will be run and operated in accordance with the rules of the Norfolk & Western Railway Company.

Trainographs and duplex gauges will be located on the first car and at five-car intervals throughout the train. The contact gauges to register predetermined pressure in the brake cylinders will be installed on the first, 17th, 34th and 50th cars.

The control of the train will be observed and when an opportunity offers thermal tests will be made of the wheels, especially at Elliston, which is at the foot of the Allegheny grade. Notes will be made as to whether stops are properly made at desired points; whether the shocks which occur, if any, as a result of brake operation, are unusual or excessive, and whether the operation of either type of triple valve has any effect, adverse or beneficial, upon the operation of the other type of triple valve. Emergency applications will be made at suitable points and notes made as to whether any undesired release of brakes occurs following emergency applications.

Locomotive Lubrication

A Review of Early Practice and Comments on the Hydrostatic vs. Force Feed or Power Lubricators

By W. E. SYMONS

From the construction of the first steam locomotive in 1828 to the present day, one of the most important and at times perplexing problems has been that of lubricating the valves, cylinders and driving journals, other parts requiring lubrication having been less troublesome to designers or operating officers.

A review of some incidents in connection with this question is of more than passing interest.

At the annual meeting of the American Railway Master Mechanics' Association at Louisville, Ky., 1871 (52 years ago), a committee reported that from the best information obtainable from various sources, local conditions more than any individual merit of one oil over another seemed to govern. The consensus of opinion however, was expressed in the following language of committee's report on the subject:

"The general preference seems to be largely in favor of the use of lard oil mixed with a proportion of earth oil of a gravity of 28 degrees or 29 degrees to keep it thin in winter. For fast passenger trains pure lard oil seems to afford the best results, while on slow freight trains heavy earth oils are equally successful, unless a journal should become over-heated, when it is almost impossible to cool it with any of the different varieties of earth oils.

"In lubricating cylinders, pure refined tallow, clear of acids, meets with most favor; although some prefer tallow oil in winter. Lard oil is also used to some extent for this purpose, securing complete immunity from dirt in pistons or in cylinder heads, and from the wasting of valve yokes so often seen when tallow is exclusively used. A system of steam lubrication is also mentioned as being successful. A sufficient quantity of steam to lubricate the valves and cylinders, but not enough to accelerate the speed on descending grades. A little tallow is used with it, about one pound per one hundred miles. By use of this process brass rings are reported as wearing ¼ inch in 44,652 miles' running.

"The use of plumbago, mixed with tallow, is highly recommended for cylinders, as giving them a smooth glazed surface, and is perhaps the best lubricant known for this purpose, as by its use the disastrous effects of acids in tallow are entirely avoided. There has heretofore been a difficulty experienced in procuring this material entirely free from foreign substances, especially quartz. But this objection to its use is obviated in the manufacture of chemically pure plumbago.

"Plumbago grease for journals has not been extensively enough used to warrant as yet its general adoption, although parties using it heartily commend it as a journal lubricator of great value and a sure prevention of hot boxes. The manufacturers claim that it is infusible and practically indestructible, as neither a heat of 4,000 degrees nor a temperature of 50 degrees below zero, nor acid, nor greases, affect its condition. Experience proves that it forms on wearing surfaces a coating of extraordinary smoothness, and from its pasty condition is not liable to be thrown from the journal boxes, as are the oils ordinarily used for this purpose."

The next report by a committee of this body was submitted at their annual meeting in New York in 1875, which indicated that 15 master mechanics out of 20 used tallow exclusively for valves and cylinders, one line got 105 miles per *quart*, one 117 per *quart*, D. L. & W. 56 miles per *quart*, C. R. R. of N. J. 27 miles per *quart*, and P. W. & B. 22 miles per *quart*. The general verdict was in favor of pure tallow, and general preference favorable to cab tubes to supply oil to cylinders and valves.

For guides and journals petroleum mixed with lard, fish oil or tallow gave good results.

The subject was a slippery one, and the Railway Master Mechanics' Association equally so in committing themselves to any standard practice or lubricant, as only 26 members responded to inquiries sent to all lines for information on the subject.

In 1877 and 1878 tallow, tallow oil or lard fed through pipes from engine cab to valves and cylinders was still accepted practice.

In 1884 quite a lengthy report was submitted to the association, and tallow was still principally used with valve oil, a mixture of petroleum and animal fats was used by some, and the sight feed lubricator was first introduced.

In 1894 the question of lubricating oils was gone into at much length, many methods or devices and character of lubricants were mentioned. What was then considered a long run, 135 to 140 miles for passenger engines, called for special facilities for lubrication, particularly axle boxes, the Lake Shore and the Erie provided an extra fount of oil on running board with feed pipes to journals, and by this means the long runs were successfully made.

The use of Galena oils were at this time being quite generally used in this country, while in England a mixture of 33 per cent rape with 66 per cent mineral oils was pretty generally used, although some used 50 per cent of each.

At the Master Mechanics' Convention of 1900, much time was devoted to the question of lubrication, journal bearings, lubricators, etc., but there was no mention of grease for driving journals. The hydrostatic lubricator being the only device considered for delivering oil to valves and cylinders.

At the Master Mechanics' Convention in 1904 the use of grease for driving journals and crank pins was practically endorsed, and the "Elvin" grease cellar was the only device mentioned suitable for proper application of grease to driving journals.

High steam pressures, with correspondingly high temperature, was giving considerable trouble in lubrication of valves and cylinders, and the experts of hydrostatic lubricator companies, with those of the Galena Oil Company were actively engaged in trying to find a satisfactory solution of this problem.

Many new improvements were made both in character of cylinder oils and devices employed in its application with varying degrees of success, for while some roads reported complete cures, others using the same remedy claimed to find little or no relief.

Force Feed or Mechanical Lubricator

The successful use of force feed lubricators in stationary and marine practice throughout the country and on locomotives abroad prompted resort to the method in this country, and after considerable experimental work and much expense, there was in use on locomotives in the United States on January 1, 1919, something over 700, or about 1.12 per cent of the total engines equipped with the Schlacks system of force feed power driven pumps, and since that time the force feed pump for the lubrication of valves and cylinders has made considerable progress, and from all reports bids fair to meet with more general favor in future.

Advantages of Power Driven Lubricators

One of the advantages, if not the principal feature in favor of the use of force feed in place of the hydrostatic sight feed lubricator is its dependability to supply a pre-

determined quantity of oil based on the service and capacity of the engine, and the absolute impossibility to overfeed or waste any oil by the engine crew, hostlers, or any one else, and while there may be times when the accessability of the hydrostatic lubricator to the engine crew for this purpose may serve to good advantage, its abuse or the great waste of oil from this source not only more than offsets this apparent benefit, but more than fully justifies the use of a device or system by which such waste is an absolute impossibility, and for this reason it looks reasonable to assume it will come into more general use.

It may be of interest to mention that on one of the best managed western lines, with an average mileage on valve oil of about 35 to the pint, and life of cylinder packing rings of less than 20,000 miles, with hydrostatic sight feed lubricators, several heavy passenger engines were equipped with force feed lubricators only and the mileage was increased to above 80 per pint of oil, and cylinder packing rings were inspected that had made from 90,000 to 115,000 miles and found to be in perfect condition with so little wear as to be hardly noticeable. This is certainly a splendid endorsement of the force feed lubricator, and incidentally may be a criticism of the method of using the hydrostatic, from which such poor results were obtained.

From the foregoing one may form an idea of the great waste of lubricating oil and the consequent expense resulting therefrom when small saturated steam engines with a pressure of 150 to 180 pounds, and cylinders 15 to 18 inches in diameter, used or wasted one *quart* of oil in a distance of 117 to as low as 22 miles, when today, engines many times in size and capacity, make more miles per *pint* of oil than was then secured from one *quart*. To those who originated and introduced the feature of "Cost of Lubrication per Thousand Miles," credit is due for having saved the railways many hundreds of million dollars on this item alone, while the principle extended to and resulted in great economies in other directions.

Comparison of Hydrostatic vs. Mechanical Lubricator

Hydrostatic Lubricator

The arguments *in favor* of the hydrostatic lubricator are as follows:

1.—It is extremely simple, cannot get out of order, and has a low first and maintenance cost.

2.—It is always under the control of the enginemen who can check the oil feed, and can increase this in emergencies, as in the case of a "blowing" engine, or where the use of bad water must be offset by increase in cylinder oil feed.

3.—The oil is fed on the basis of time and not on the basis of piston travel, thus giving the maximum amount of oil when it is most necessary, i.e., under heavy load.

4.—As the oil is introduced into the cylinder by steam, or, as it is sometimes expressed, is carried by steam, good distribution of the oil is effected even when drifting. Furthermore, the steam in the lubricator keeps the oil fluid at all times and assists in its atomization.

5.—As it feeds on the time principle it always gives an ample supply of oil for the locomotive when starting.

6.—There being nothing to get out of order it is absolutely dependable and will function so long as there is steam in the boiler and the lubricator valves are open.

7.—It forms less carbon than the mechanical lubricator.

Against Hydrostatic Lubricator

The argument *against* the hydrostatic, as offered by the proponents of the mechanical lubricator, are as follows:

1.—The feed is based on time only, regardless of the rubbing speed of the piston, which is manifestly illogical.

2.—The rate of speed is controlled by the engineman which is a disadvantage, since few enginemen have a real conception of the science of lubrication and the general tendency is to overfeed, causing carbon formation.

3.—It must be shut off when standing to avoid oil wastage and carbonization. Where frequent short stops are made—as in local train service—the enginemen are not likely to shut off the oil feed and, therefore, the consumption of lubricant is greater than is necessary or desirable.

4.—When running under full load with full boiler pressure in the cylinders the oil feed is simply due to the small column of water in the lubricator. On the other hand, when drifting, the oil is fed under heavy pressure. This is bound to cause irregularity in feed and excess quantities of oil being supplied to the cylinders when drifting, tending to produce objectionable carbon. Probably one reason that the advocates of the hydrostatic lubricator claim that lubrication under heavy load is difficult, is due to this very fact, since at such times the pressure on either side of the lubricator is almost equalized, and there is very little force left to carry the oil through the long length of pipe extending from the engine cab to the steam line.

Mechanical Lubricator

The arguments used by the advocates of the mechanical lubricator are:

1.—The feed is based upon the actual rubbing speed of the moving parts.

2.—The rate of feed being set by the shop crew this cannot be varied according to the whim of the enginemen. Therefore the oil may be set at the minimum rate to give satisfactory lubrication, thus reducing the formation of carbon. Furthermore, there can be no waste of lubricant by the enginemen forgetting, or not troubling, to shut off the lubricant when the locomotive is standing.

3.—Oil is only fed when required. It starts as the locomotive starts and stops as the locomotive stops.

4.—It is absolutely fool-proof, is adjusted and filled by the Maintenance Division, taking the responsibility out of the hands of the enginemen and putting it in the hands of those who are responsible for the maintaining of the locomotive in proper running condition.

5.—Regardless of whether the engine is running or drifting, the oil is always fed under the same pressure, thereby assuring an absolutely regular and constant oil feed.

6.—Because of various reasons outlined above, it is capable of giving efficient lubrication with far less oil than the hydrostatic.

7.—It contains no explosive pressure.

8.—It gives less for the enginemen to attend to.

9.—If desired, it can be so driven from the valve action as to deliver an increased quantity of oil for heavy loads, i.e., long cut-off.

10.—It can be used to lubricate the axle boxes also if desired.

Against Mechanical Lubricator

The arguments *against* the mechanical lubricator are as follows:

1.—It is costly both to install and maintain.

2.—It gets out of order easily.

3.—It is out of control of the enginemen, who furthermore cannot be certain that the oil is feeding.

4.—It is affected by temperature changes.

5.—It does not take care of the Westinghouse Air pump.

6.—If the locomotive drive on the lubricator fails, lubrication is not obtainable.

7.—It is disliked by many enginemen.

8.—It will not lubricate a blowing engine, since the feed cannot be increased by the driver.

9.—It will not take care of heavy load conditions.

10.—Wear in the drive mechanism affects the rate of feed.

11.—When drifting the oil is not carried into the cylinder by a steam jet, as in the case of the hydrostatic lubricator, and for the same reason, there is no real atomization of the oil in such cases. This forms carbon faster.

12.—It feeds the oil on the wrong principle, since more is required at low speeds and heavy loads than at high speeds and light loads.

So much for the arguments for and against each type of device.

We will not discuss further the arguments advocating the two different lubricators, but we believe that a few comments might be desirable as to those *against each*. We will, therefore, take them up herewith by number.

Against the Hydrostatic

1.—This is open to discussion. It must be admitted, however, that the rubbing speed is an important factor in all lubrication analysis.

2.—On the majority of railroads the engineer has a fixed oil allowance he cannot exceed, and a competent man will vary his rate of feed to meet the requirements. The point is open to discussion, however, and it is worth noting that the advocates of the mechanical lubricator place just as much stress on its being desirable to take the control of the lubrication out of the enginemen's hands, as the supporters of the hydrostatic lubricator place on its being an advantage. And each side has excellent arguments and can cite experience to prove its contentions. Unquestionably a first class engineer can get excellent results with a hydrostatic lubricator. The problem is rather as to whether the mechanical lubricator does not give greater economy with the *average* engineer.

3.—Granted.

4.—We have been unable to obtain any logical disproof of this argument.

Against the Mechanical

1.—It is admitted that the installation is costly. Also the maintenance may be conceded to be more expensive than that of the hydrostatic lubricator, but it is not "costly" where a properly designed and applied mechanical lubricator is used. Furthermore, the saving in lubrication which is obtainable more than offsets the increase in maintenance cost.

2.—This does not seem to be substantiated. It may perhaps have been true of the spring-operated-valve type of mechanical lubricator, although it is not of the present valveless design.

3.—As indicated in No. 2 under the discussion of arguments against the hydrostatic lubricator, this point is open to debate. Both sides have excellent arguments to support their contentions. Judging from the class of enginemen in many foreign fields, and the general attitude of enginemen in this country on this question, we would be inclined to think that control of lubrication should primarily rest with mechanical officers. Furthermore, it must be admitted that the human element after all is more likely to fail than is the mechanical. However, it is possible to control and regulate the oil feed from the cab if desired, and this is done in several European countries, although we have as yet no details of the devices employed. Also, many lubricators, not cab controlled, have tell-tales visible from the engineer's seat.

4.—Advocates of the mechanical lubricator claim that in a properly designed device—and each one insists that he makes or uses it—temperature changes have no effect on the oil feed. This would seem a subject for impartial investigation.

5.—Granted—but this does not seem to be a really serious objection.

6.—Granted—but also of relatively little importance. The hydrostatic lubricator would fail if its steam lead broke.

7.—This is really not a valid argument against the ability of the lubricator itself. It must be admitted that many enginemen are antagonistic, and endeavor to discredit the operation of the mechanical lubricator.

8.—Granted. There is no question but what there are cases where trouble develops on a run, and repairs cannot be made until the end of the run is reached. In such a situation there can be no question of the desirability of being able to supply an excess of oil to the mechanically defective cylinder. The question here is rather as to whether this disadvantage is not more than offset by the various advantages of the mechanical lubricator.

9.—If we are to accept as correct the teachings of our own engineers, as set forth in our Educational Training Course, we cannot admit that a heavy load is a more difficult position to lubricate with a mechanical lubricator, although we can see where it might be with a hydrostatic, for the reasons set forth in argument 4 against the hydrostatic lubricator. It would appear to us that the delivery of the proper amount of oil for both light and heavy loads is largely a question of setting. If a mechanical lubricator is adjusted to give the proper amount of lubricant per stroke, when the engine is working under a heavy load, it will unquestionably continue to do so under a light load.

10.—This does not seem to be based on any really sound foundation.

11.—We cannot feel that this point with regard to drifting is really supported. Modern practice very generally involves the use of drifting valves, and the steam thus admitted is in excess of the small amount which is introduced by a hydrostatic lubricator. Thus there *is* a medium to carry the oil admitted during drifting by a mechanical lubricator. Furthermore, it must be granted that owing to unbalanced pressure, the hydrostatic lubricator is likely to feed excess oil at such times. With a properly set mechanical lubricator less carbon will form. Where carbon trouble has been experienced with the latter device, it has usually been due to the instrument having been set on the basis of the rate necessary for the hydrostatic lubricator, which would be too much.

12.—This is open to discussion. We are inclined to disagree.

The above sums up, as concisely and impartially as possible, the data which we have been able to collect from various sources. We are inclined to believe that the mechanical locomotive lubricator is growing in interest with the advent of the improved types, and with the introduction of larger cylinders and higher steam pressures and temperatures.

Fuel Association Convention Papers

A well-founded program of interest to railroad officers and coal operators has been arranged for the annual meeting of the International Railway Fuel Association to be held at the Hotel Winton, Cleveland, Ohio, May 21-24. The following list gives the addresses, papers and reports which have already been definitely arranged. The opening address will be made by Julius Kruttschnitt, chairman of the executive committee of the Southern Pacific. Mr. Kruttschnitt has chosen as his subject Railway Fuel—

There are some who, through a process of consistent and logical reasoning, possibly prompted by the observations of the performance of force feed lubricators, are staunch supporters of the force feed method of lubrication and consider it superior to the hydrostatic principle. On the other hand, there are others, possibly representing a majority, who are equally staunch advocates of the hydrostatic method of lubrication. It may be, in many instances, the knowledge of unsatisfactory results with force feed lubricators in the earlier designs has, in a measure, shaped the attitude they have taken although they support the stand they have taken as advocates of the hydrostatic principle of lubrication by a logical process of reasoning.

The Committee, in canvassing the force feed lubrication situation, was advised that there were twelve railroads upon which the method of force feed lubrication was being investigated. A questionnaire was prepared with the purpose of covering some of the more pertinent factors relative to the two systems of lubrication and sent to the roads reported as having force feed installations.

It may be of interest to indicate the roads replying to the questionnaire showing the type and number of force feed lubricators they have in service and the location of the oil delivery:

Road	Lubricators System	No.	No. of Feeds	Location of Oil Delivery
C. & E. I.	Schlack	1	2	In outside steam pipe.
C. & N. W.	Schlack	—	2	In outside steam pipe.
	Madison-Kipp	—	2	In outside steam pipe.
	Nathan Mfg. Co.'s Friedmann	—	6	In outside steam pipe.
C. M. & St. P.	Schlack	1	2	In outside steam pipe.
D. L. & W.	Madison-Kipp	—	2	In main steam pipe.
	McCord	—	2	In main steam pipe.
Grand Trunk	Had Intensifore and Wakefield. Both now removed.		5	To outside steam pipe and to top of cylinder barrel.
N. & W.	Schlack	1	2	In top of cylinder bar.
	McCord	5	2	To mallet LP cyls.
	Goodfellow	1	4	To steam pipe.
	Madison-Kipp	1	2	To steam pipe.
Pennsylvania	Nathan Mfg. Co.'s Friedmann Type FSS	1	6	To steam pipe.
	Nathan Mfg. Co.'s Friedmann Type DV	1	6	To steam pipe.
	McCord	50	2	To steam pipe.
	Schlack	50	2	To steam pipe.
Southern Pacific	Nathan Mfg. Co.	—	4	One to each steam pipe and top of cylinder barrel.
Union Pacific	Madison-Kipp	—	4	One to each outside steam pipe and top of each cyl. barrel.
	Schlack	—	4	Bottom of cylinder.
Erie	Madison-Kipp	—	4 & 6	Bottom of cylinder.
	Nathan Mfg. Co.	—	4	Bottom of cylinder.

The three systems of force feed lubrication that have received any measure of recognition by the railroads are the Nathan Mfg. Co's Friedmann, Madison-Kipp, Schlack (formerly identified as the McCord).

The Committee, in its efforts to gather together information that would be of value to the Association in the consideration of this subject has not met with the success that should attend the pursuit of a subject of such primary importance and which should undoubtedly lead to improvement in valve and cylinder packing ring service. The roads that have taken the initial step in the application of one or more force feed lubricators have shown a disposition to co-operate with your Committee by supplying such information as they have relative to the lubricators but they were not in position to furnish data that would be indicative of the relative results obtained from force feed and hydrostatic lubrication, doubtlessly due to the limited experience they have had with force feed lubricators and their records of the relative performance of hydrostatic and force feed systems were not complete.

In the discussion on this subject Messrs. Hardin, Brooks, Kearney and Lanning participated, but nothing constructive or of real value was brought out, although Mr. Brooks mentioned an increase of 30 per cent in mileage made with force feed lubrication, and 35 to 50 per cent in life of packing rings.

The subject will be continued with suggestions from the committee as to conducting tests of different types of lubricants.

Taking Up Driving and Truck Wheel Lateral

From the information that the committee was able to obtain no patented devices have been applied in quantity and little is known of the service, application and maintenance costs.

In the absence of conclusive information as to the reliability of the various patented devices for taking up driving and truck wheel lateral, and in view of the satisfaction evidently obtained either by pouring or pinning bronze liners to the face of the driving box or wheel hub, as evidenced by their almost universal use, it would seem that until some better device is developed that the liner poured or pinned to the driving box face or driving wheel hub is the best means of taking up lateral.

Driving wheel lateral: Discussed by Messrs. Gray, Chambers, Cadington, Goodwin and Wiggins.

Repeated calls for information as to some of the special methods such as pressure from grease back of loose liner, but not a word or even a grunt from any one. The remarks were entirely on conventional types of liners, brass, cast iron, steel cast, plate, etc.

If the members' aversion to speaking on the subject may be taken as a measure of progress then there is nothing new in this line.

In all walks of life are men, who under certain circumstances are brave as a lion while the same individual may at times be as meek as a lamb.

A most striking example of this dual personality was revealed in repeated attempts on the part of Messrs. Bentley, Giles, Kearney, Oviatt, Chairman Coleman and a few others to get the members to respond. Numerous members with a storehouse full of important information, who when outside the hall are usually afflicted with chronic "Furore Liquendi" seemed to have become ossified when in the convention hall, and no amount of persuasion would induce them to get up on their hind legs and talk.

Tractive Power Mallet Locomotives

The formula now in general use for the simple locomotive gives only a close approximation of the actual tractive power for such a locomotive, because simple locomotives vary in maximum cut-off, back pressure, internal friction, etc. The same reasons that justify the adoption of a uniform formula for tractive power for simple locomotives, also justify the adoption of a uniform formula for the Mallet locomotive, but it must be understood that the formula adopted will give only a close approximation of the actual performance of the Mallet locomotive.

There are two general types of Mallet locomotives:

1st, the four cylinder simple, and

2nd, the four cylinder compound.

C = Diameter of high pressure or simple cylinder in inches.

c = Diameter of low pressure cylinder in inches.